Feasibility Study of Real-time Neutron Measurement using a Diamond detector for the LIBRA experiment

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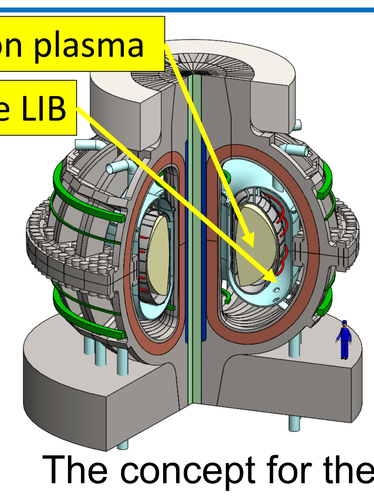
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Introduction

The Liquid Immersion Blanket (LIB)

- The liquid immersion blanket (LIB) has been proposed for the fusion power plant (FPP), such as the ARC reactor [1].
- FLiBe molten salt has functions to cool the reactor, shield magnets, and breed Tritium.
- The LIB surrounds the vacuum vessel and plasma in the FPP.

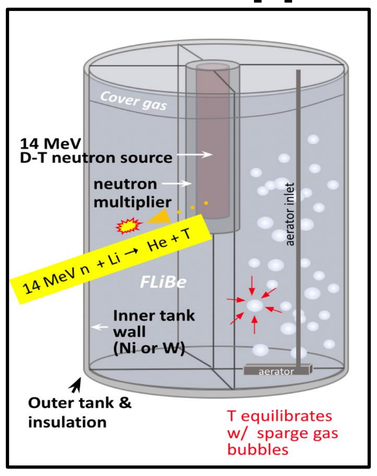


The concept for the ARC reactor [1]

The LIBRA Experiment [2]

Liquid Immersion Blanket Robust Accountancy

- The LIBRA experiment will demonstrate the achievability of the LIB concept. The DT fusion neutron irradiation to FLiBe is planned.
- **Characterization of the fusion neutron source** is critical for the LIBRA experiment for the evaluation of the Tritium accountancy.
- Neutron measurement is also critical in the LIBRA experiment. **Diamond detectors** are planned to be used for real-time monitoring of neutrons.



The LIBRA experiment [2]

Objectives

This study aims to:

① characterize the DT fusion neutron source.

② validate the feasibility of the neutron measurement by using a diamond detector.

Measurement by Imaging Plate and Activation foil

The characterization for the DT neutron source was conducted using the activation foil and the imaging plate (IP).

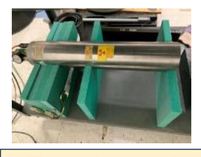
Method

1. DT fusion neutrons are irradiated to the activation foil (Al) around the neutron source.
2. After the irradiation, the activation foils are attached to the IP(exposing process). The IP has a sensitivity for a γ-ray and a β-ray.
3. After the exposing process, the IP is scanned by the scanner to obtain the image.

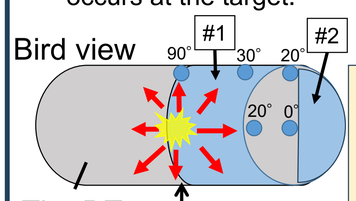
Table: Nuclear reaction between neutron and aluminum [3].

Nuclear reaction	Half-life time of generated nuclei	Eth (MeV)
$^{27}Al(n,\alpha)^{24}Na$	15.0 h	3.3
$^{27}Al(n,p)^{27}Mg$	9.46 m	1.9

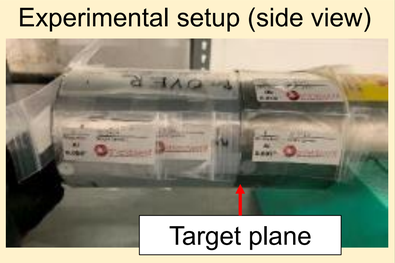
1 DT fusion reaction (D + T → He + n) occurs at the target.



The DT fusion neutron source

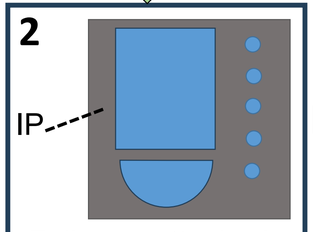


Experimental setup (side view)



Large Al foils (#1,#2) and small Al foils are set on the surface.

2



Foils are attached to the IP.

3

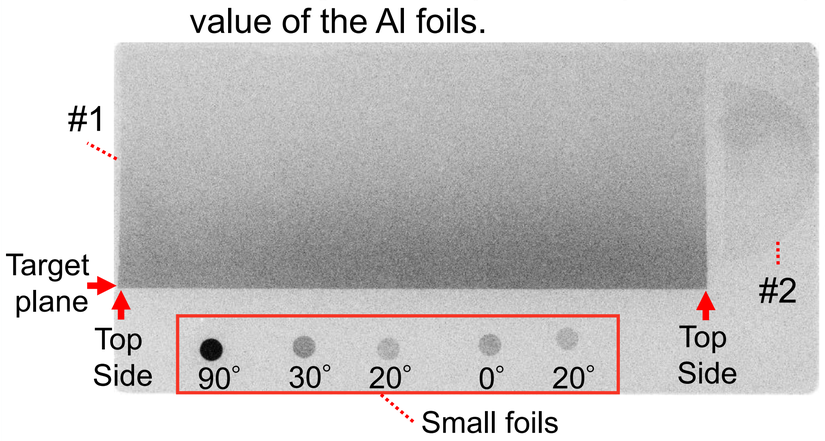


The IP scanner (Typhoon FLA 7000 (GE))

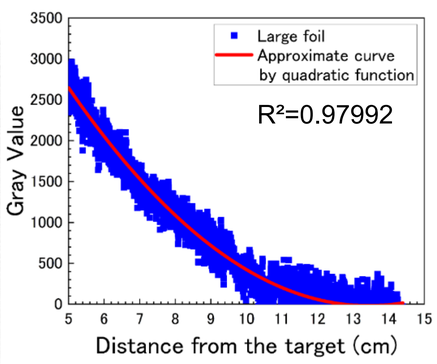
The image can be obtained by the IP scanner.

Result

The IP can detect a radiation from the activated Al foils.
The image intensity at each point corresponds to the total decay value of the Al foils.



The IP image from the activated Al foils



The image gray value along the incident particle direction in #1(large Al foil).

The comprehensive neutron distribution around the neutron source was obtained by using the Al activation foils and the IP.

Measurement by Diamond detector

Principle of a diamond detector

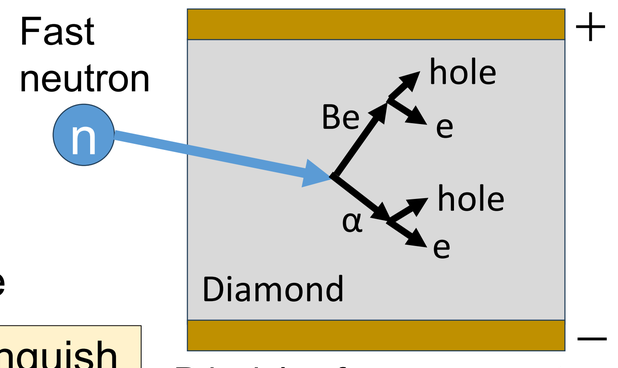
A diamond detector can detect fast neutrons by using a $^{12}C(n, \alpha)^{9}Be$ reaction.

$^{12}C + n \rightarrow {}^{9}Be + {}^{4}He$

($E_{threshold}$: 6.181 MeV)

- High band gap
- High radiation resistance
- High temperature resistance

The diamond detector can distinguish a DT neutron and a DD neutron.



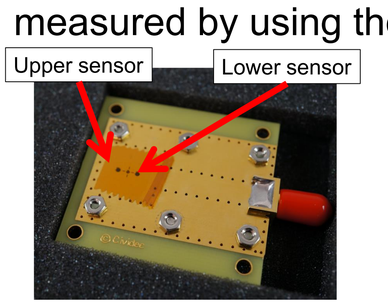
Principle of measurement by a diamond detector

Experiment and Simulation

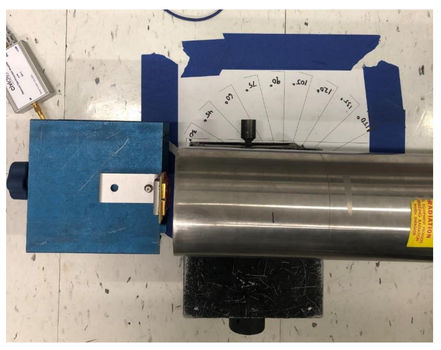
The accelerator-based neutron source generally shows the anisotropic neutron distribution [4].

Experiment

The angular distribution of neutron around the neutron source was measured by using the sCVD diamond detector (made by Cividec).



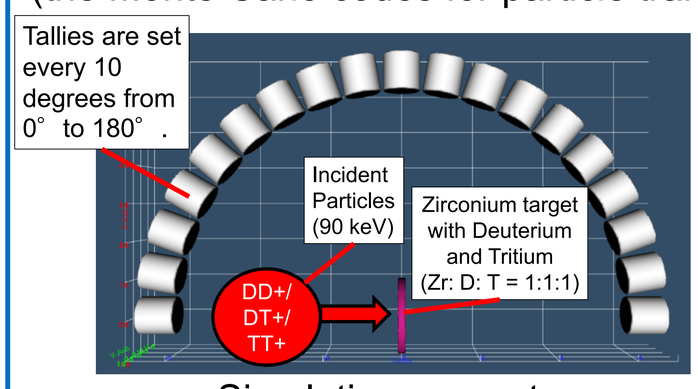
Fast-Neutron Diamond detector with two sensors.(made by Cividec)



DT neutron was measured for 30 minutes at every 15 degrees from 30° to 180° degrees. The distance from the target was 15 cm.

Simulation

The simplified simulation was also performed by using the PHITS code (the Monte Carlo codes for particle transport simulation)[6].

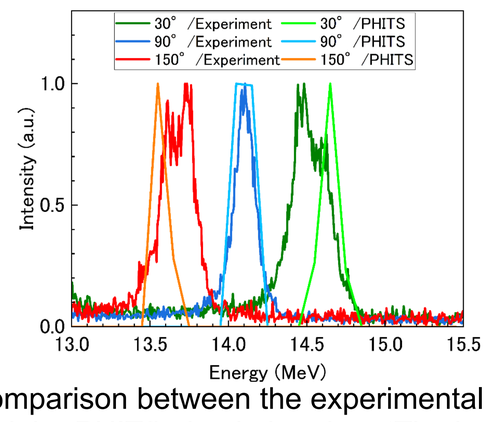


Simulation geometry

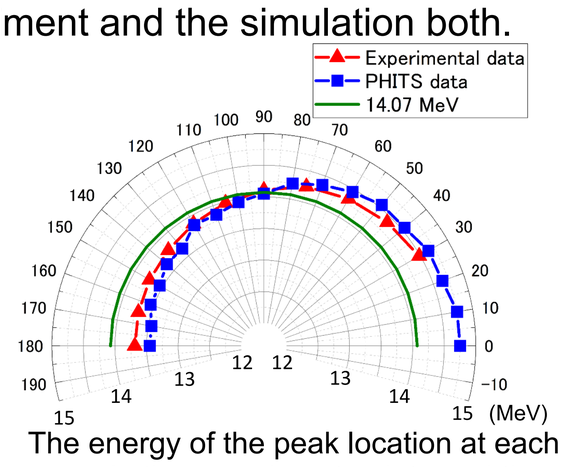
In this study, collisions between Deuterium and/or Tritium were simulated.
CP2020 ACE format charged particle transport library [7] was used for the simulation.

Result

The angular dependency of the DT neutron can be observed in the experiment and the simulation both.



Comparison between the experimental data and the PHITS simulation data (The incident particle direction is set as 0 degrees).



The energy of the peak location at each angle in the experimental data and the PHITS simulation data.

The real-time monitoring by using the diamond detector can be performed. The tendencies of the angular energy distribution are same between the experiment and simulation.

Conclusion

◆The characterization of the DT neutron source was successfully carried out using the method of using the imaging plate and Aluminum activation foils.

◆The angular distribution of DT neutrons can be measured by using the diamond detector in real-time. The tendency of the distribution highly agreed with the simulation result.

◆Further testing in the experimental system containing FLiBe is necessary.

[1] B. N. Sorbom et al., Fusion Engineering and Design, vol.100, 378-405, 2015 [2] Sara E. Ferry et al., Fusion Science and Technology, vol.79, 13-35, 2023 [3] S. Kunieda et al., Proceedings the 2015 Symposium on Nuclear Data, 41-46, 2015 [4] M. Sharma et al., Nuclear Inst. and Methods in Physics Research, A 911, 37-44, 2018 [5] https://cividec.at/index.php?module=public.product&idProduct=5&scr=0 (accessed June 19 2023) [6] T. Sato et al., Journal of Nuclear Science and Technology, vol.50, 913-923, 2013 [7] https://nucleardata.lanl.gov/ace/cp2020/ (accessed June 19 2023)